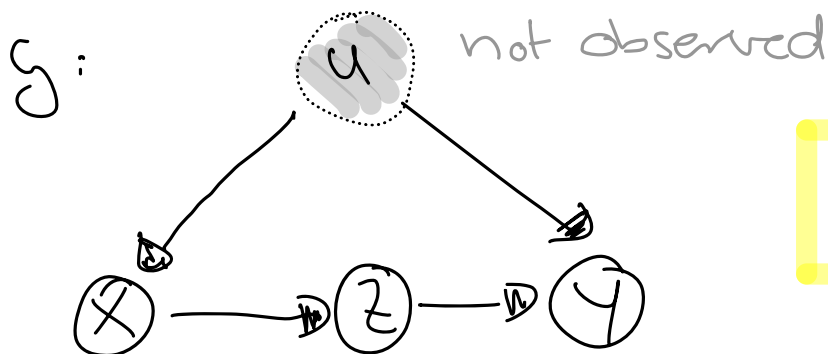
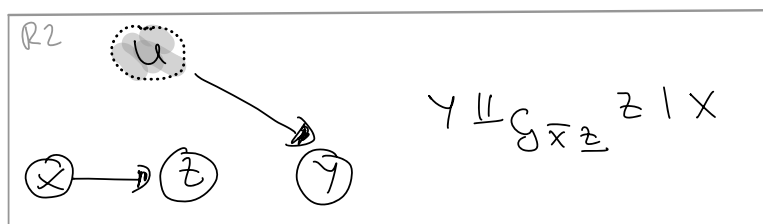


Front-Door Adjustment (derivation)

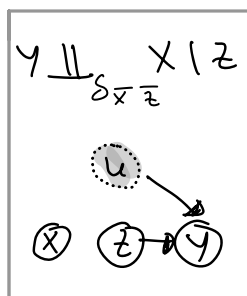


$$p(y | do(x)) = ?$$

$$p(y | do(x)) = \sum_z p(y | z, do(x)) p(z | do(x))$$



$$\begin{aligned} (R2) \\ &= \sum_z p(y | \text{do}(z), \text{do}(x)) p(z | do(x)) \end{aligned}$$



$$\begin{aligned} (R3) \\ &= \sum_z p(y | do(z)) p(z | x) \end{aligned}$$

$$\begin{aligned} R2 \\ &= p(z | x) \mid z \perp\!\!\!\perp_{G_x} x \end{aligned}$$

$$= \sum_z p(z | x) \boxed{p(y | do(z))}$$

X fulfills back-door criterion

X from original do-intervention

$$= \sum_z p(z | x) \left[\sum_{x'} p(y | z, x') p(x') \right]$$